

# Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

## WEEK-3

Name: D. Venya Sri

Roll no: 2520030531

### INSERT SAMPLE DOCUMENTS

Organization: DROPATHI VENYA SR... PROJECT: Project 0

Data Explorer

Cluster0 > bookflow > book\_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

|   | book_metadata                  |               |                               |                    |                      |          |
|---|--------------------------------|---------------|-------------------------------|--------------------|----------------------|----------|
|   | _id ObjectId                   | book_id Int32 | title String                  | author String      | published_year Int32 | category |
| 1 | ObjectId('6a9641be2df7513...') | 101           | "The Martian"                 | "Andy Weir"        | 2021                 | "Scienc" |
| 2 | ObjectId('6a9641e62df7513...') | 102           | "Dune"                        | "Frank Herbert"    | 2022                 | "Scienc" |
| 3 | ObjectId('6a9641fb2df7513...') | 103           | "Clean Code"                  | "Robert C. Martin" | 2019                 | "Progra" |
| 4 | ObjectId('6a9642192df7513...') | 104           | "Artificial Intelligence ..." | "John Smith"       | 2023                 | "Artifi" |
| 5 | ObjectId('6a9642262df7513...') | 105           | "Data Structures"             | "Mark Allen"       | 2024                 | "Comput" |

### COMPARISON OPERATOR \$gt

Organization: DROPATHI VENYA SR... PROJECT: Project 0

Data Explorer

Cluster0 > bookflow > book\_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

```
{
  "reviews.rating": {
    "$gt": 4
  }
}
```

[Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

### Output

| book_metadata |                                |               |                               |                    |                      |          |
|---------------|--------------------------------|---------------|-------------------------------|--------------------|----------------------|----------|
|               | _id ObjectId                   | book_id Int32 | title String                  | author String      | published_year Int32 | category |
| 1             | ObjectId('6a9641be2df7513...') | 101           | "The Martian"                 | "Andy Weir"        | 2021                 | "Scienc" |
| 2             | ObjectId('6a9641e62df7513...') | 102           | "Dune"                        | "Frank Herbert"    | 2022                 | "Scienc" |
| 3             | ObjectId('6a9641fb2df7513...') | 103           | "Clean Code"                  | "Robert C. Martin" | 2019                 | "Progra" |
| 4             | ObjectId('6a9642192df7513...') | 104           | "Artificial Intelligence ..." | "John Smith"       | 2023                 | "Artifi" |
| 5             | ObjectId('6a9642262df7513...') | 105           | "Data Structures"             | "Mark Allen"       | 2024                 | "Comput" |

USING \$IN

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

ORGANIZATION

DROPATHI VENYA SR...

PROJECT

Project 0

Welcome

book\_metadata

+

Cluster0 > bookflow > book\_metadata

Documents 5

Aggregations

Schema

Indexes 1

Validation

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

Output

| book_metadata |                              |               |                               |                    |                      |          |
|---------------|------------------------------|---------------|-------------------------------|--------------------|----------------------|----------|
|               | _id ObjectId                 | book_id Int32 | title String                  | author String      | published_year Int32 | category |
| 1             | ObjectId('6a9641be2df7513... | 101           | "The Martian"                 | "Andy Weir"        | 2021                 | "Scienc  |
| 2             | ObjectId('6a9641e62df7513... | 102           | "Dune"                        | "Frank Herbert"    | 2022                 | "Scienc  |
| 3             | ObjectId('6a9641fb2df7513... | 103           | "Clean Code"                  | "Robert C. Martin" | 2019                 | "Progra  |
| 4             | ObjectId('6a9642192df7513... | 104           | "Artificial Intelligence ..." | "John Smith"       | 2023                 | "Artifi  |
| 5             | ObjectId('6a9642262df7513... | 105           | "Data Structures"             | "Mark Allen"       | 2024                 | "Comput  |

REGEX QUERY

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

ORGANIZATION

DROPATHI VENYA SR...

PROJECT

Project 0

Welcome

book\_metadata

+

Cluster0 > bookflow > book\_metadata

Documents 5

Aggregations

Schema

Indexes 1

Validation

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

Output

| book_metadata |                              |               |                               |                 |                      |          |
|---------------|------------------------------|---------------|-------------------------------|-----------------|----------------------|----------|
|               | _id ObjectId                 | book_id Int32 | title String                  | author String   | published_year Int32 | category |
| 1             | ObjectId('6a9641be2df7513... | 101           | "The Martian"                 | "Andy Weir"     | 2021                 | "Scienc  |
| 2             | ObjectId('6a9641e62df7513... | 102           | "Dune"                        | "Frank Herbert" | 2022                 | "Scienc  |
| 3             | ObjectId('6a9642192df7513... | 104           | "Artificial Intelligence ..." | "John Smith"    | 2023                 | "Artifi  |
| 4             | ObjectId('6a9642262df7513... | 105           | "Data Structures"             | "Mark Allen"    | 2024                 | "Comput  |

\$MATCH

**Data Explorer**

My Queries

Data Modeling

**CLUSTERS (1)**

Search clusters

Cluster0

ORGANIZATION

DROPATHI VENYA SR...

PROJECT

Project 0

Welcome

book\_metadata

Cluster0 > bookflow > book\_metadata

Documents 5

Aggregations

Schema

Indexes 1

Validation

{ "published\_year": { "\$gt": 2020 } }

Output

| book_metadata |                                |               |                               |                 |                      |          |
|---------------|--------------------------------|---------------|-------------------------------|-----------------|----------------------|----------|
|               | _id ObjectId                   | book_id Int32 | title String                  | author String   | published_year Int32 | category |
| 1             | ObjectId('6a9641be2df7513...') | 101           | "The Martian"                 | "Andy Weir"     | 2021                 | "Scienc" |
| 2             | ObjectId('6a9641e62df7513...') | 102           | "Dune"                        | "Frank Herbert" | 2022                 | "Scienc" |
| 3             | ObjectId('6a9642192df7513...') | 104           | "Artificial Intelligence ..." | "John Smith"    | 2023                 | "Artifi" |
| 4             | ObjectId('6a9642262df7513...') | 105           | "Data Structures"             | "Mark Allen"    | 2024                 | "Comput" |

\$UNWIND

Stage 1

\$unwind

1

"\$reviews"

Output

Output preview after [\\$unwind](#) stage (Sample of 10 documents)

id: ObjectId('6a7c4317e71fafc1f3d8ef44')

book\_id: 104

title: "Artificial Intelligence Basics"

author: "John Smith"

published\_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file\_size: "4.2 MB"

member\_id: 104

\_id: ObjectId('6a7c4317e71fafc1f3d8ef44')

book\_id: 104

title: "Artificial Intelligence Basics"

author: "John Smith"

published\_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file\_size: "4.2 MB"

member\_id: 104

\$GROUP

ORGANIZATION

DROPATHI VENYA SR...

PROJECT

Project 0

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

bookflow

book\_metadata

local

sample\_mflix

book\_metadata

Cluster0 > bookflow > book\_metadata

Documents 5

Aggregations

Schema

Indexes 1

Sunwind

\$group

Untitled - modified

SAVE

+ CREATE NEW

EXPOR

Stage 2 \$group

```

1 {
2   "$group": {
3     "_id": "$title",
4     "average_rating": {
5       "$avg": "$reviews.rating"
6     },
7     "total_reviews": {
8       "$sum": 1
9     }
10  }
11 }
```

## Output

Output preview after \$group stage (Sample of 5 documents)

|                                                                        |                                                                              |
|------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <pre> _id: "Data Structures" average_rating: 4 total_reviews: 2 </pre> | <pre> _id: "Dune" average_rating: 4.6666666666666666 total_reviews: 3 </pre> |
|------------------------------------------------------------------------|------------------------------------------------------------------------------|

## \$SORT

ORGANIZATION

DROPATHI VENYA SR...

PROJECT

Project 0

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

bookflow

book\_metadata

local

sample\_mflix

book\_metadata

Cluster0 > bookflow > book\_metadata

Documents 5

Aggregations

Schema

Sunwind

\$group

\$sort

Untitled - modified

SAVE

CREATE NEW

Stage 3

\$sort

1 {

2 |"\$sort": {

3 "average\_rating": -1

4 }

5 }

## Output

Output preview after \$sort stage (Sample of 10 documents)

\_id: ObjectId('6a7c4317e71fafc1f3d8ef44')

book\_id: 104

title: "Artificial Intelligence Basics"

author: "John Smith"

published\_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file\_size: "4.2 MB"

member\_id: 104

\_id: ObjectId('6a7c4317e71fafc1f3d8ef44')

book\_id: 104

title: "Artificial Intelligence Basics"

author: "John Smith"

published\_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file\_size: "4.2 MB"

member\_id: 104

Create index text

ORGANIZATION DROPATHI VENYA SR... PROJECT Project 0

**Data Explorer**

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- bookflow
- book\_metadata**
- local
- unmnlx mfliv

book\_metadata

Cluster0 > bookflow > book\_metadata

Documents 5 Aggregations Schema **Indexes 2** Validation

Create Index Refresh

Looking for search indexes?  
These indexes can be created and viewed under [Search and Vector Search](#).

| Name & Definition     | Type      | Size    | Usage                      | Properties | Status |
|-----------------------|-----------|---------|----------------------------|------------|--------|
| _id_                  | REGULAR 1 | 36.9 kB | 10 (since Tue Sep 01 2026) | UNIQUE 1   | READY  |
| reviews.comments_text | TEXT 1    | 20.5 kB | 0 (since Tue Sep 01 2026)  |            | READY  |

## SEARCH USING TEXT INDEX

ORGANIZATION DROPATHI VENYA SR... PROJECT Project 0

**Data Explorer**

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb

book\_metadata

Cluster0 > bookflow > book\_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

Generate query

```
{
  "$text": {
    "$search": "excellent"
  }
}
```

## Output

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

|   | book_metadata | _id ObjectId                 | book_id Int32 | title String                  | author String      | published_year Int32 |
|---|---------------|------------------------------|---------------|-------------------------------|--------------------|----------------------|
| 1 |               | ObjectId('6a7c4317e71fafc... | 101           | "The Martian"                 | "Andy Weir"        | 2021                 |
| 2 |               | ObjectId('6a7c4317e71fafc... | 102           | "Dune"                        | "Frank Herbert"    | 2022                 |
| 3 |               | ObjectId('6a7c4317e71fafc... | 103           | "Clean Code"                  | "Robert C. Martin" | 2019                 |
| 4 |               | ObjectId('6a7c4317e71fafc... | 105           | "Data Structures"             | "Mark Allen"       | 2024                 |
| 5 |               | ObjectId('6a7c4317e71fafc... | 104           | "Artificial Intelligence ..." | "John Smith"       | 2023                 |

All Good

## SEARCH FOR A SPECIFIC KEYWORD

ORGANIZATION

DROPATHI VENYA SR...

PROJECT

Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

book\_metadata

Cluster0 > bookflow > book\_metadata

Documents 5

Aggregations

Schema

Indexes 2

Validation

```

{
  "$text": {
    "$search": "\"science fiction\""
  }
}

```

## Output

ADD DATA

BULK

EXPORT CODE

25

1 - 2 of 2

↺

↻

↷

≡

{ }

⌕

book\_metadata

|   | _id ObjectId                 | book_id Int32 | title String  | author String   | published_year Int32 |  |
|---|------------------------------|---------------|---------------|-----------------|----------------------|--|
| 1 | ObjectId('6a7c4317e71fafc... | 102           | "Dune"        | "Frank Herbert" | 2022                 |  |
| 2 | ObjectId('6a7c4317e71fafc... | 101           | "The Martian" | "Andy Weir"     | 2021                 |  |

## USE explain()

ORGANIZATION

DROPATHI VENYA SR...

PROJECT

Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

book\_metadata

Cluster0 > bookflow > book\_metadata

Documents 5

Aggregations

Schema

Indexes

```

{
  "$text": {
    "$search": "excellent"
  }
}

```

## Output



ADD DATA

BULK

EXPORT CODE

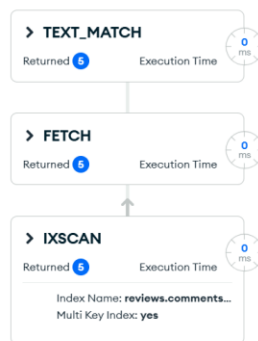
251 - 5 of 5

| book_metadata |                              |               |                              |                    |                      |  |
|---------------|------------------------------|---------------|------------------------------|--------------------|----------------------|--|
|               | _id ObjectId                 | book_id Int32 | title String                 | author String      | published_year Int32 |  |
| 1             | ObjectId('6a7c4317e71fafc... | 101           | "The Martian"                | "Andy Weir"        | 2021                 |  |
| 2             | ObjectId('6a7c4317e71fafc... | 102           | "Dune"                       | "Frank Herbert"    | 2022                 |  |
| 3             | ObjectId('6a7c4317e71fafc... | 103           | "Clean Code"                 | "Robert C. Martin" | 2019                 |  |
| 4             | ObjectId('6a7c4317e71fafc... | 105           | "Data Structures"            | "Mark Allen"       | 2024                 |  |
| 5             | ObjectId('6a7c4317e71fafc... | 104           | "Artificial Intelligence ... | "John Smith"       | 2023                 |  |

## Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual TreeRaw Output



### Query Performance Summary

- 5 documents returned
  - 5 documents examined
  - 0 ms execution time
  - Is not sorted in memory
  - 5 index keys examined
- Query used the following index:
- \_fts (text)
  - \_ftsx ↑

Close